Campus Wi-Fi Congestion Predictor

Project ID : 25
Phase II

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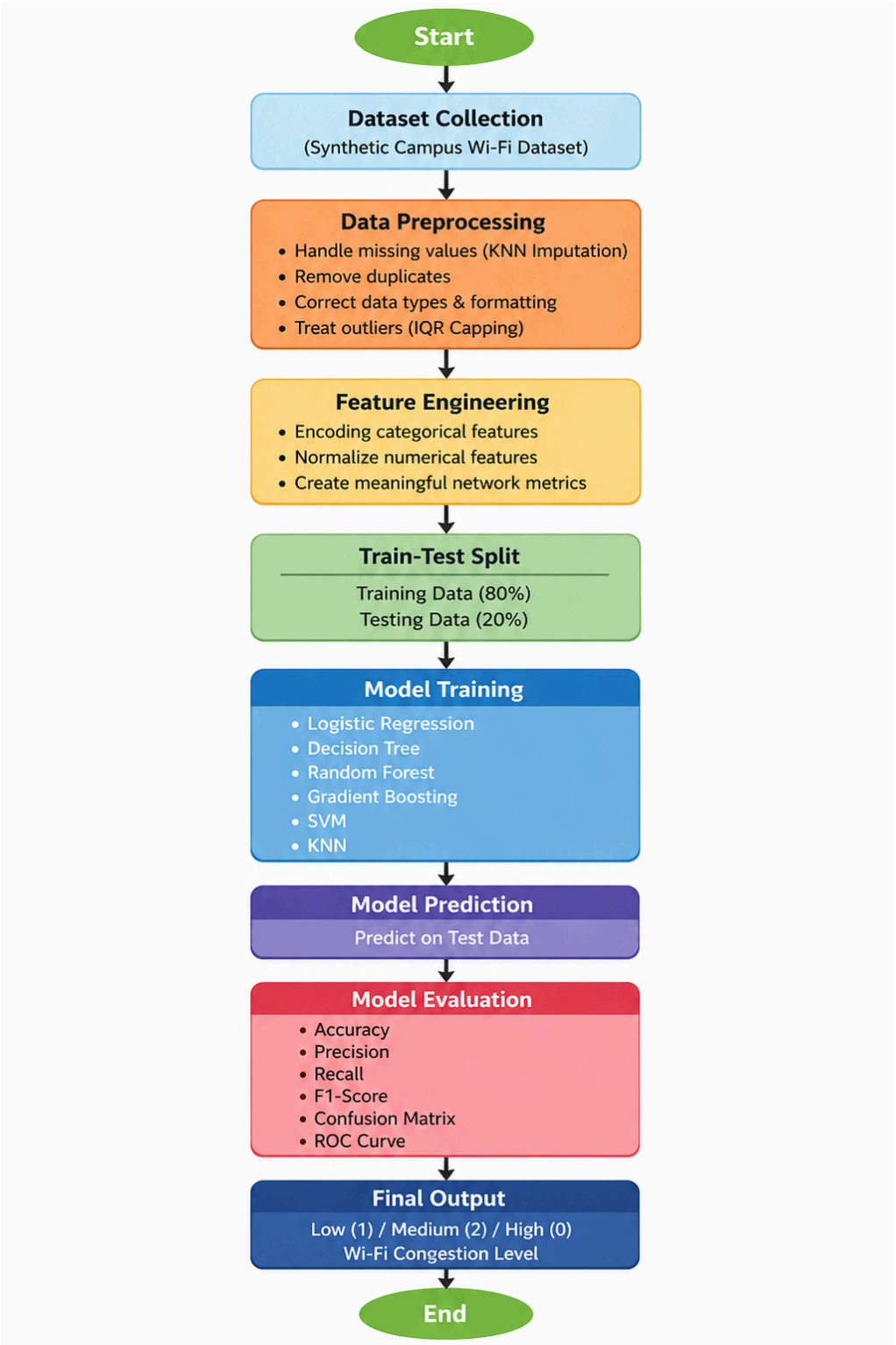
Problem Description :

The Campus Wi-Fi Network Congestion Predictor is a Python-based machine learning application developed to automate the detection and prediction of campus network congestion levels. It streamlines the process of collecting real-time network performance data, training predictive models, and generating congestion insights while maintaining high accuracy and computational efficiency. Using core concepts of Machine Learning such as data preprocessing, feature engineering, PCA-based dimensionality reduction, multi-class classification algorithms, model evaluation, and statistical visualization libraries, this project minimizes manual effort in network monitoring and supports data-driven infrastructure management decisions for university IT departments.

Architecture :



Models Used :

The Campus Wi-Fi Network Congestion Predictor employs six machine learning classification algorithms to predict network congestion levels as Low, Medium, or High. Each model was trained on the PCA-transformed feature set and evaluated using accuracy, precision, recall, and F1-score to identify the best performing classifier.

- **Logistic Regression** serves as the baseline linear classification model. It estimates the probability of each congestion class using a sigmoid function and is particularly useful for understanding the linear separability of the feature space before applying more complex algorithms.
- **Decision Tree Classifier** builds a tree-structured model by recursively splitting the dataset based on feature thresholds that best separate the three congestion classes. It offers high interpretability, making it easy to visualize and understand the decision rules governing congestion prediction.
- **Random Forest Classifier** is an ensemble method that constructs multiple decision trees during training and aggregates their predictions through majority voting. It emerged as the best performing model in this project, achieving an accuracy of 98.20%, owing to its robustness

against overfitting and ability to handle high-dimensional PCA-transformed features effectively.

- **Gradient Boosting Classifier** builds trees sequentially, where each new tree corrects the errors made by the previous one. This iterative error-correction approach makes it highly accurate on structured tabular data and serves as a strong competitor to Random Forest in terms of predictive performance.
- **Support Vector Machine (SVM)** identifies the optimal hyperplane that maximizes the margin between the three congestion classes in the transformed feature space. Using an RBF kernel, it handles non-linearly separable data effectively and is particularly useful when the decision boundary between Low, Medium, and High congestion is complex.
- **K-Nearest Neighbors (KNN)** classifies each record by identifying the K most similar records in the training set and assigning the majority congestion class among them. It is a non-parametric, instance-based model that requires no explicit training phase and serves as a strong benchmark for distance-based classification performance.

Evaluation Metrics Used :

To evaluate the performance of the Campus Wi-Fi Congestion Predictor, multiple evaluation metrics were used to ensure that the trained machine learning models could accurately classify network congestion into **Low, Medium, and High** categories. Since this project involves multi-class classification, using only one metric would not provide a complete understanding of model effectiveness. Therefore, a combination of performance metrics was applied to compare models and select the most reliable one.

- **Accuracy Score :** Accuracy was the primary metric used in this project to measure the overall percentage of correctly predicted congestion levels out of all predictions made. After applying SMOTE to balance the dataset, accuracy became a strong indicator of model performance and was especially useful for comparing all six classification models. It played the most important role in identifying the best-performing model, where the Random Forest Classifier achieved the highest accuracy of **98.20%**.
- **Precision :** Precision was used to measure how accurately the model predicted each congestion class, particularly High congestion cases. In the context of campus Wi-Fi management, high precision is important because it reduces false alarms, ensuring that when the system predicts severe

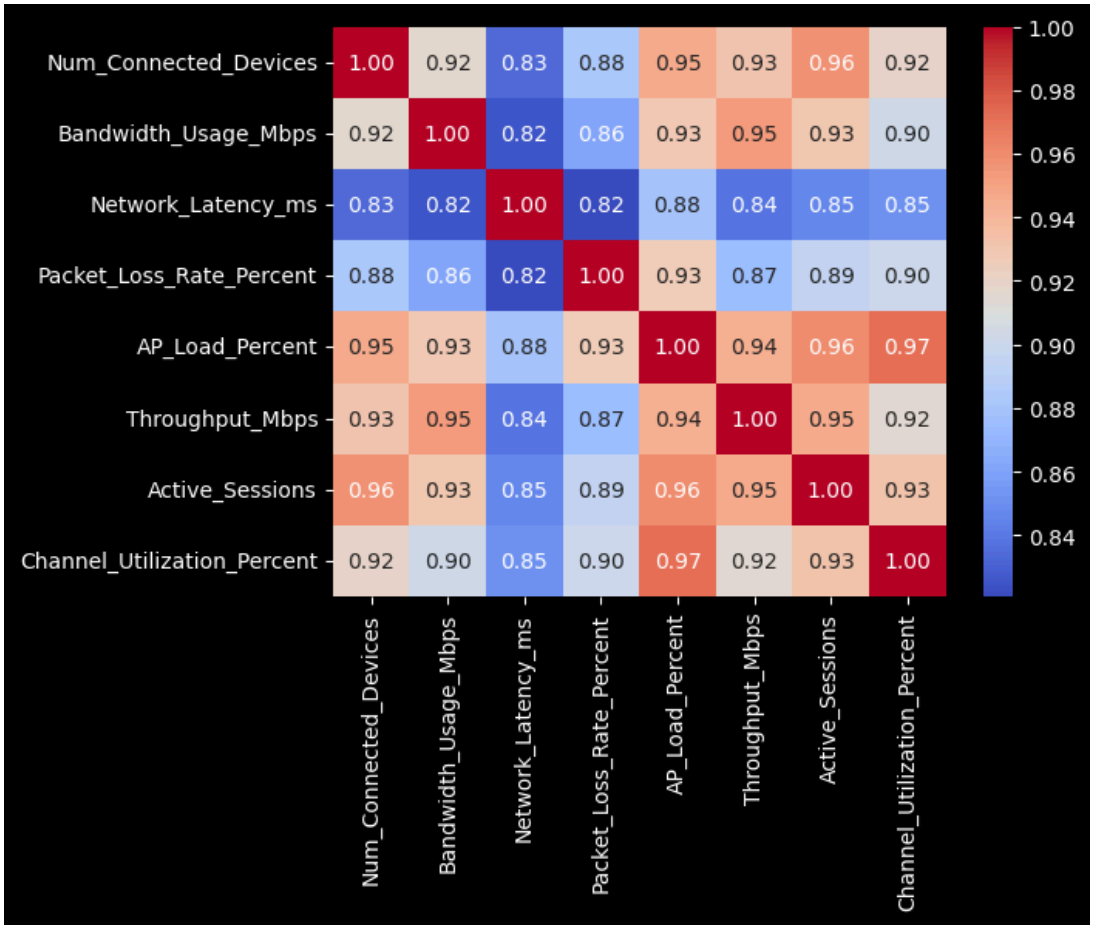
congestion, it is likely to be correct. This makes the model more practical for real-world proactive monitoring.

- **Recall :** Recall was used to evaluate how effectively the model detected actual congestion events, especially Medium and High congestion states. This metric was important because failing to detect genuine congestion could negatively impact students and staff during peak usage periods. High recall ensured that critical congestion situations were identified as early as possible.
- **F1-Score :** F1-Score was used to balance Precision and Recall, providing a more complete measure of model quality for each congestion class. Since both false positives and false negatives matter in network prediction, F1-Score helped determine which model maintained the best balance between accurate and complete predictions.
- **Classification Report :** A classification report was generated for each model to provide class-wise Precision, Recall, and F1-Score values for Low, Medium, and High congestion categories. This allowed deeper analysis beyond overall accuracy and helped identify whether a model was consistently performing well across all congestion levels.
- **Confusion Matrix :** The confusion matrix was used as a visual evaluation tool to analyze correct and incorrect predictions across all three classes. It clearly showed how often one congestion category was misclassified as another, making it valuable for understanding model weaknesses

and improving reliability. This was especially useful for checking whether High congestion was being confused with Medium or Low congestion.

- **10-Fold Cross Validation :** To ensure that the final model was not overfitting and could generalize well to unseen data, 10-Fold Cross Validation was performed. This method tested the model across multiple train-test splits and confirmed the consistency and robustness of the selected Random Forest model.
- **Final Outcome :** By combining these evaluation metrics, the project was able to compare all trained classifiers comprehensively and scientifically. Based on overall performance, stability, and class-wise prediction quality, the **Random Forest Classifier** emerged as the final model with **98.20% accuracy**, making it the most reliable solution for predicting campus Wi-Fi congestion.

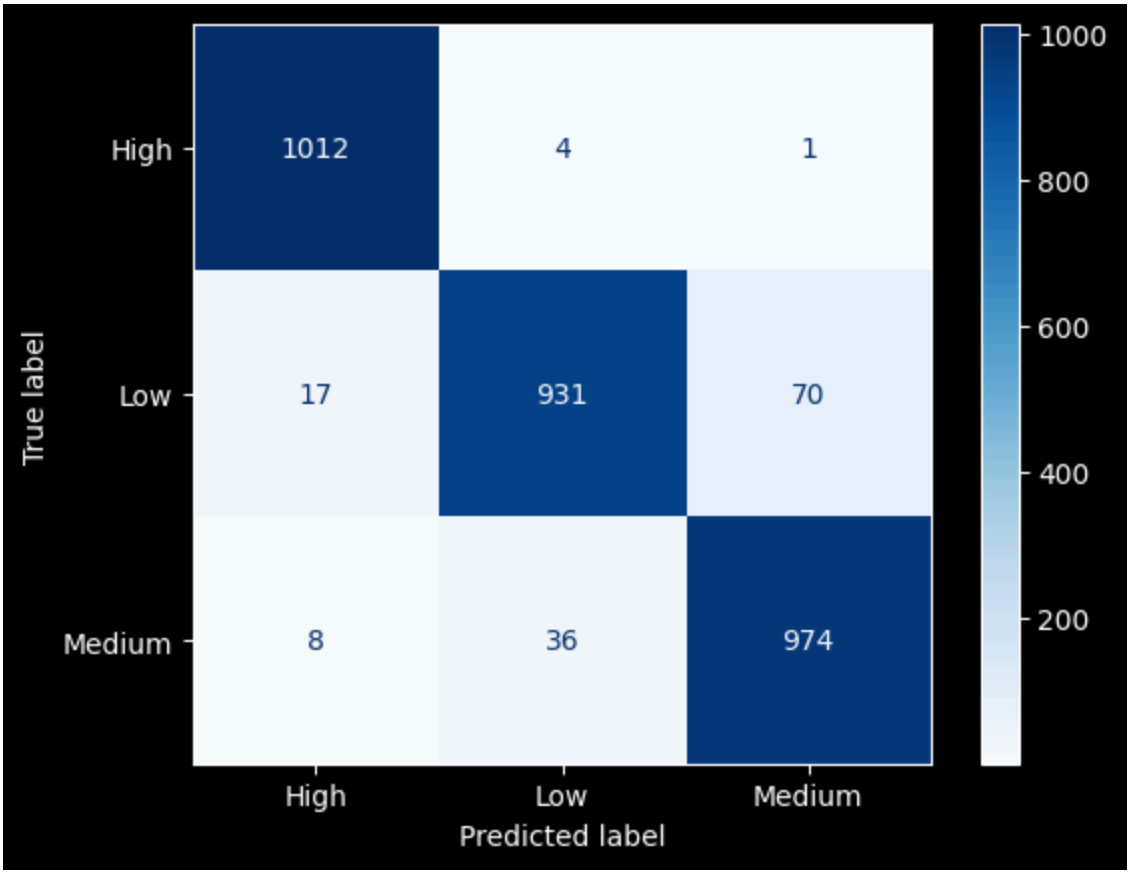
Preliminary Results And Visualization :



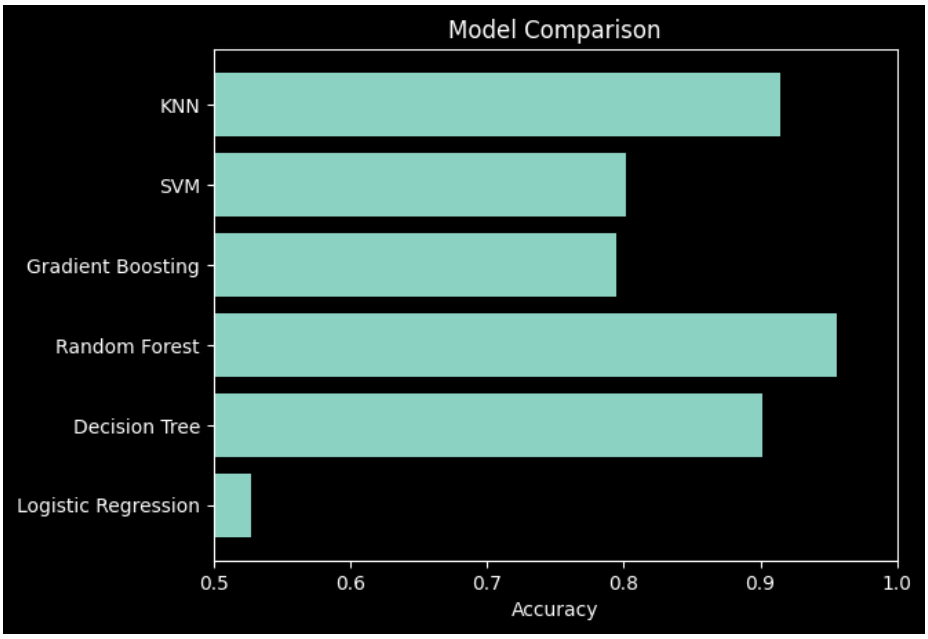
Correlation Heatmap

```
Model: Random Forest | Accuracy: 0.9555
              precision    recall  f1-score   support

        High       0.98      1.00      0.99      1017
         Low       0.96      0.91      0.94      1018
      Medium       0.93      0.96      0.94      1018

    accuracy                           0.96      3053
   macro avg       0.96      0.96      0.96      3053
weighted avg       0.96      0.96      0.96      3053
```

Best Model Evaluation Metrics



Confusion Matrix



Model Comparison

References :

Books:

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron
2. Python Machine Learning – Sebastian Raschka & Vahid Mirjalili
3. Let Us Python – Yashavant Kanetkar

Online Resources:

- Scikit-learn Official Documentation – https://scikit-learn.org/
- Pandas Official Documentation – https://pandas.pydata.org/
- NumPy Official Documentation – https://numpy.org/
- Seaborn Official Documentation – https://seaborn.pydata.org/
- Matplotlib Official Documentation – https://matplotlib.org/
- GeeksforGeeks – https://www.geeksforgeeks.org/
- Stack Overflow – https://stackoverflow.com/